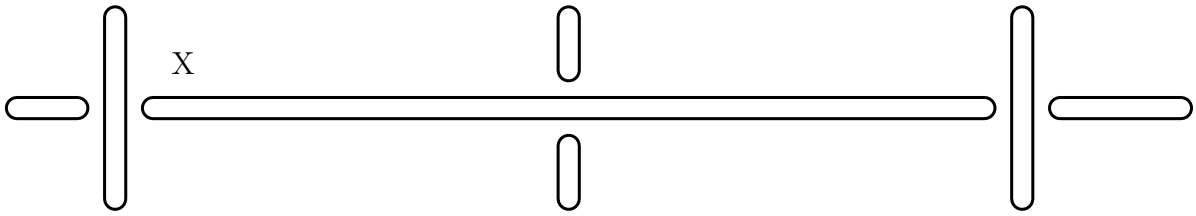
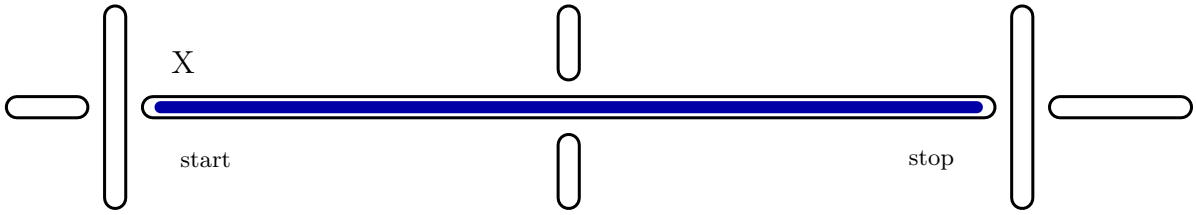
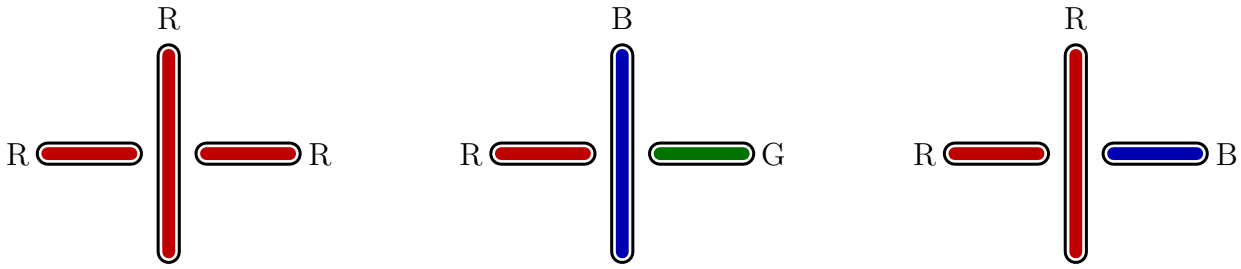
Keep cords on the table. Move them without cutting or passing strands through each other. Use red (R), blue (B), and green (G).



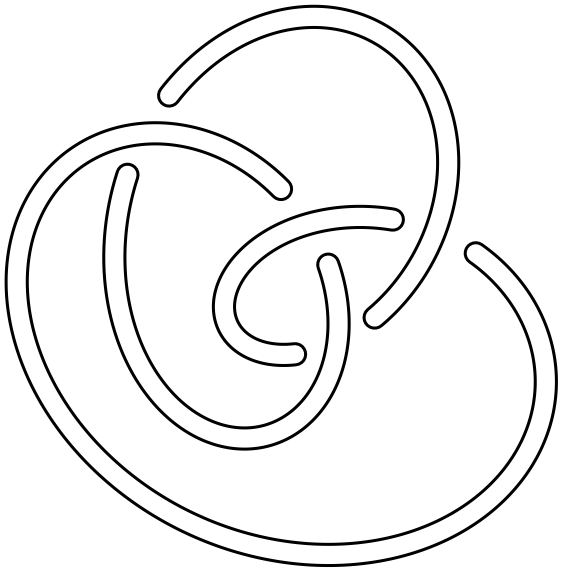
Follow X across the overpass to the next gap.



An arc runs between underpasses and continues through overpasses. A closed strand with no underpasses is one whole arc. Give each whole arc one color. At each crossing the colors must be all the same or all different.

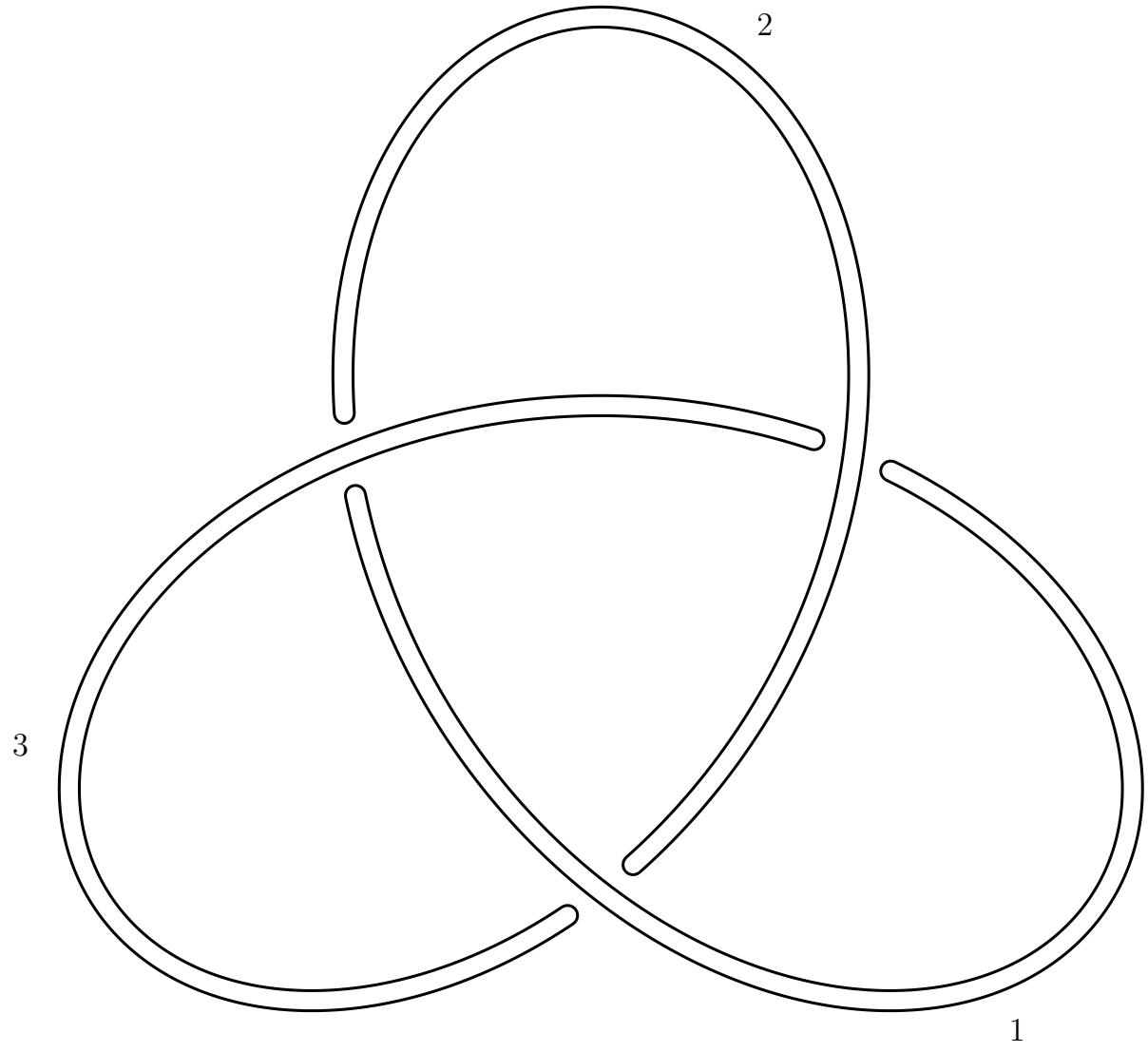


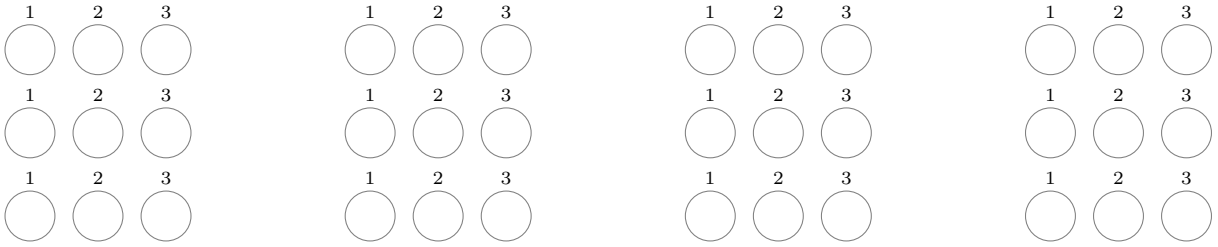
Problem 1: How many whole arcs does this loop have?



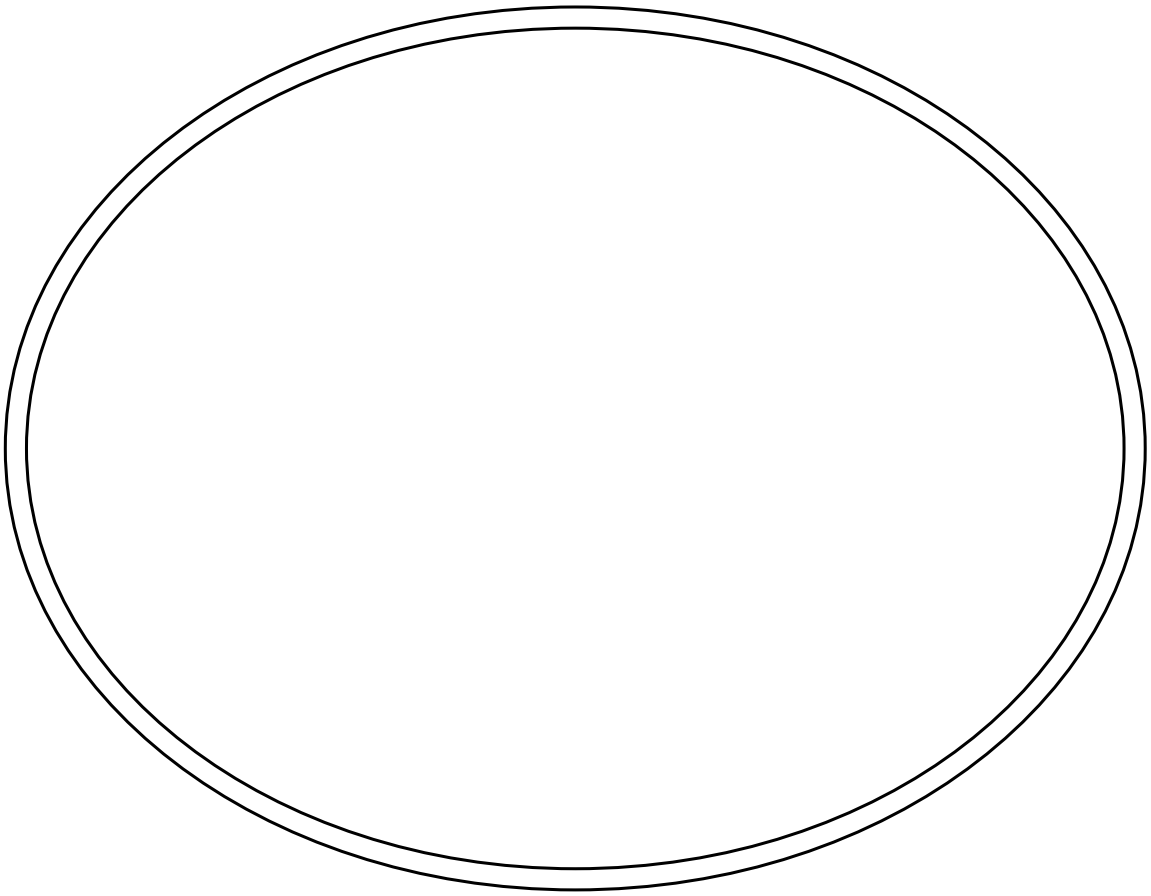
arcs: _____

Problem 2: Find every valid coloring of this loop, including one-color choices. The numbers name whole arcs; record each coloring and explain why your list is complete.



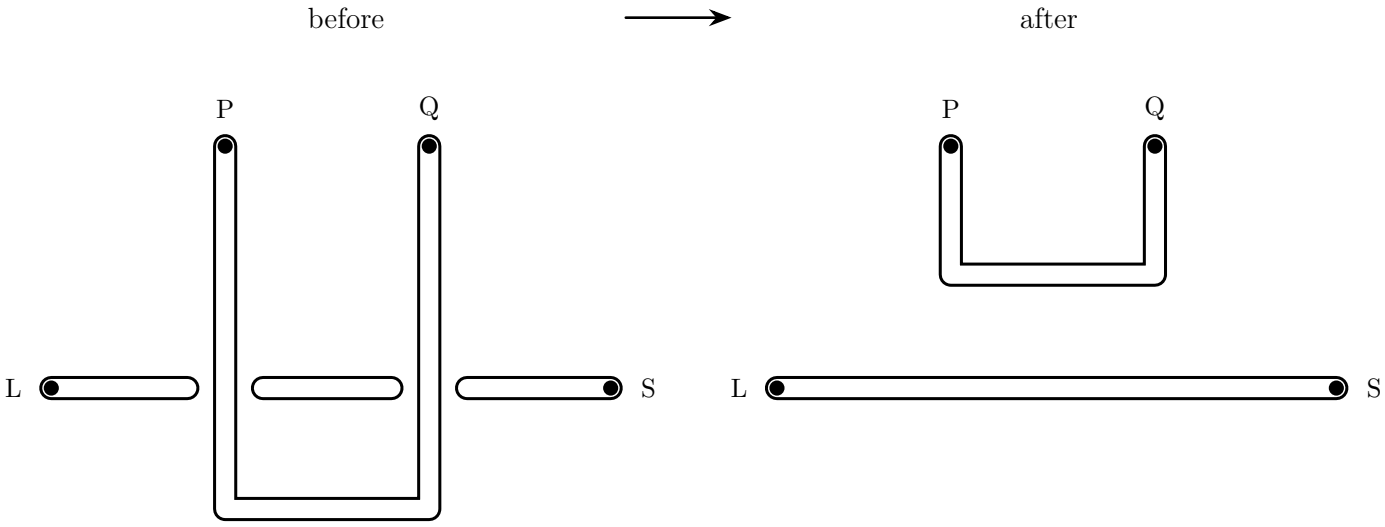


Problem 3: Count all valid colorings of this plain loop. Compare that count with the count for the three-crossing loop.



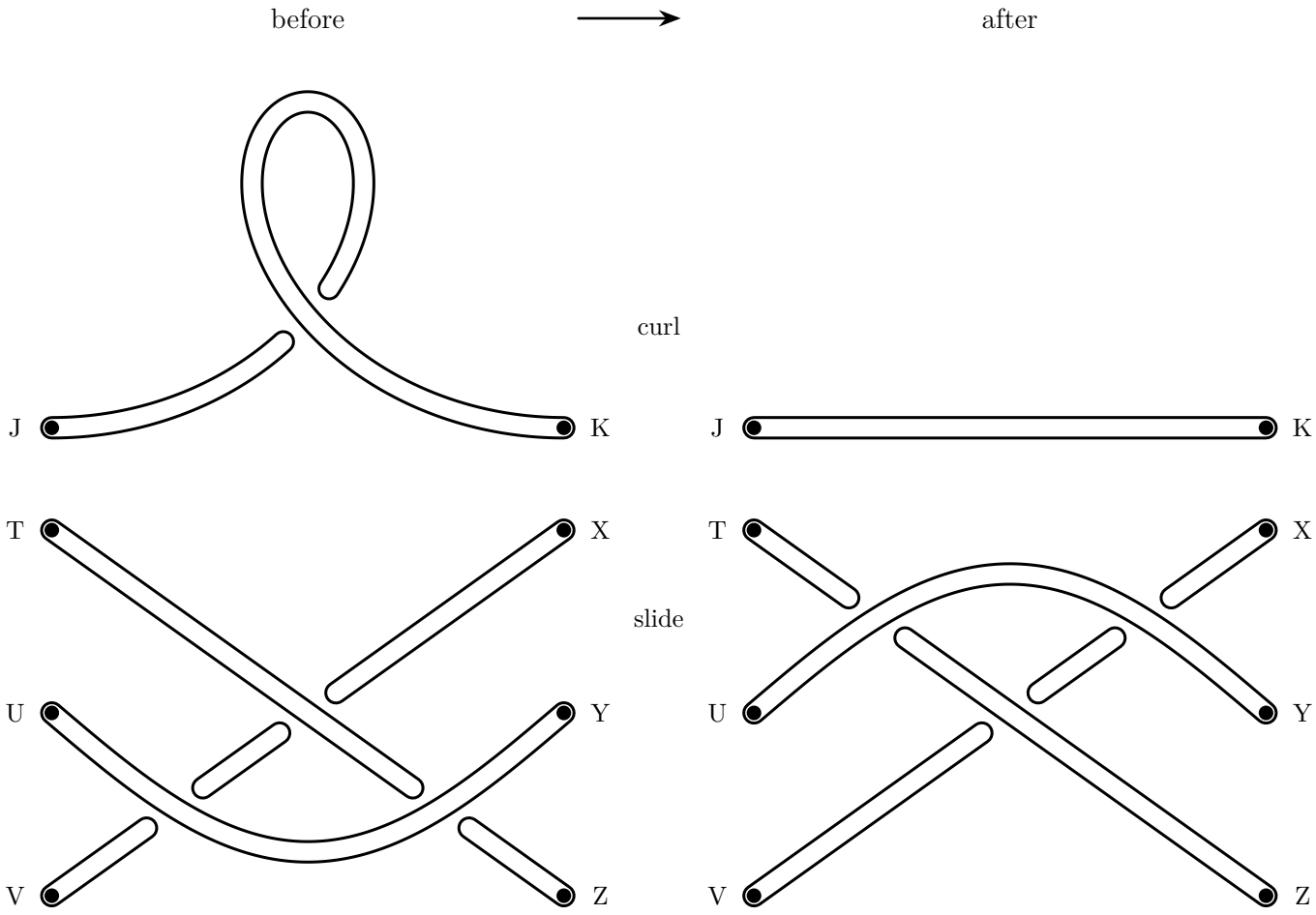
Problem 4: Try to move each of the four closed cords A, B, C and D in the tray into this plain loop shape. Separate what you managed to do from what you only suspect cannot be done.

Problem 5: An arc may stop at an end dot here. Which colors at L and P allow both pictures to be colored with matching end-dot colors? Can the same choice have more than one completion? Explain.



L	P	middle	S	Q

Problem 6: Check the curl and the slide the way you checked the move in Problem 5. Every way of moving a loop is made of curls, the move in Problem 5, and slides. Can the three-crossing loop become the plain loop?



Problem 7: Find every coloring of this loop. It cannot become a plain loop. What can equal coloring counts tell you about two loops?

